



Réseau de recherche sur les données de santé du Canada
Health Data Research Network Canada

Solving challenges in multi-regional health research

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CRDCN – Opening Doors to Data
28 October 2025

Declarations and Acknowledgements

No conflicts to declare



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Overview

- Health Data Research Network Canada
- The origins of challenges for cross-regional research
- Comparability of data (harmonized data / common data models)
- No movement of data (distributed / federated analysis)





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Health Data Research Network Canada

HDRN Canada

Health Data Research Network Canada

is a distributed pan-Canadian network that supports multi-regional data access for transformative and world-leading health data use, working to shape a data future that better supports health and health equity.



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Why is it this way in Canada?

- Constitutional Federation
- Independence
 - Legislation
 - Policies
 - Technology
- Politics



Multi-regional studies are theoretically possible, but practically infeasible (right now)

- Federation means 13 provincial/territorial (P/T) health and data systems
- Hard to understand data holdings across P/Ts to plan studies
- Data not standardized across P/Ts to enable 'apples and apples' analysis
- Data generally not moved across P/T boundaries so hard to analyze it together as one







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Solving challenges: comparability of data and no movement of data

Definitions

Information model: Describes the characteristics and relationships of information for certain key real-world entities (e.g., “hospitalization”). It does not define how the information is stored or what technology is used.

Data model: Establishes how the data element is stored in a specific system. A data model always complies with an information model, explicitly or implicitly.

Common data model: A data model agreed upon by a large group of researchers with the aim of reusing it across multiple studies sharing a common purpose.

Distributed analysis: Statistical analysis of a distributed data source. The analysis aims to protect local data privacy and does not rely on exchanging line-level data outside of local sites

Federated analysis: A type of distributed analysis using federated data. Federated data share the same data model, which simplifies analysis. Federated analysis can also mean automation of technical connections across secure environments.



Solving Challenges: Implementation of a Common Data Model

- Harmonizing requires deep local knowledge – data generation, idiosyncrasies, and changes over time
- Requires excellent documentation
- Takes a lot of time *but only needs to be done once*

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Solving Challenges: Federated Data Analysis

- Distributed / Federated analysis is the best way to analyze data without having to move them
- Statistically – very possible for many types of analyses (and an area of needed methods development)
- When done well, produces results that are essentially identical to pooled analysis
- Can be done in a privacy sensitive way and in a way that does not require high levels of trust of other secure environments.
- Federation automates the information sharing
- Getting this right will also make international research possible



Potential for OMOP Common Data Model + Federated Data Analysis

- Simplifying Production of Required Analytic Code
 - CDM provides federated data with the same data model
 - Not limited to a particular CDM, as long as it's a common model
- Supporting Rigorous "as if pooled" Analysis
 - Linear and logistic regression and survival analysis, with or without weights
 - Further development and testing underway



Summing up

- Canada is a federation and has commitments to Indigenous data sovereignty
- This creates (surmountable) challenges for research using health data not available through CRDCN (StatCan) or CIHI
- Harmonized data and common data models are up-front investments that can have large impact, if the harmonization follows common / generalizable approaches (supporting construct / content validity)
- Distributed analysis / data federation can get around data sharing issues *and can support analyses that are equal to pooled.*
- All creates new opportunities for analyses both within and across regions, and and internationally.



Thank you!

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